

1 Uncertainty and Dimensional Analysis



Source / 來源

- 選修物理(一)課堂講義 (4623R52) 教用版，第1章 不確定度，§ 1-1– § 1-4，pp. 6–17
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 1: Measurement
- Young & Freedman, *University Physics with Modern Physics*, Ch 1: Units, Physical Quantities, and Vectors

Examples

Core Ideas — Measurement and Uncertainty

An **error** is the difference between a measured value and the true value. In most experiments the true value is unknown, so the International Organization for Standardization uses **measurement uncertainty** to quantify the spread or lack of certainty in a result. A smaller uncertainty indicates a more reliable measurement.

For repeated measurements of X :

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$SD = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}, \quad u_A(X) = \frac{SD}{\sqrt{n}}$$

Type-B uncertainty uses instrument resolution or published information. With no better information and a uniform distribution,

$$u_B(X) = \frac{\text{smallest scale division}}{2\sqrt{3}}$$

Independent A- and B-type contributions combine in quadrature:

$$u(X) = \sqrt{u_A(X)^2 + u_B(X)^2}$$

Report the result as $X \pm u(X)$. Keep at most two significant figures in $u(X)$, and round the best estimate to the same decimal place.

Significant Figures

Scientific notation, $a \times 10^n$ with $1 \leq a < 10$, makes the number of significant figures unambiguous. For example, $302.06 = 3.0206 \times 10^2$ has five significant figures, $0.00205 = 2.05 \times 10^{-3}$ has three, $240.00 = 2.4000 \times 10^2$ has five, and $36\,000 = 3.60 \times 10^4$ has three when written with the shown precision.

Example 1 — Reporting a Repeated Time Measurement (*p. 8*)

A stopwatch has a smallest scale division of 0.01 s. A steel ball is dropped from 1.50 m and the fall time is measured five times: 0.58, 0.57, 0.61, 0.63, and 0.58 s. How should the measurement result be reported?

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 Solution

Example 2 — Comparing Two Balances (*p. 9*)

Atai uses a balance with a 0.1 kg smallest division; Xiaoyu uses one with a 1 kg smallest division. They measure the same fruit basket four times.

Measurement	Atai (kg)	Xiaoyu (kg)
1	16.77	16.8
2	16.75	
3	16.77	16.7
4	16.78	16.8
Mean	16.7675	16.775
SD	0.01258	0.05

Find each reported result and decide which measurement is more trustworthy.

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 Solution

Try It 1 — Pendulum Period

Ming measures a pendulum period four times: 1.3, 1.2, 1.3, and 1.4 s. The calculator gives $\bar{x} = 1.3000$ s and $SD = 0.08165$ s. The stopwatch resolution is 0.1 s. Which is the correct standard-uncertainty report?

(A) 1.300 ± 0.029 s

(B) 1.300 ± 0.041 s

(C) 1.300 ± 0.050 s

(D) 1.300 ± 0.081 s

(E) 1.300 ± 0.10 s

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 Solution

Core Ideas — Propagation of Uncertainty

For independent quantities x and y :

$$z = ax \pm by \quad \Longrightarrow \quad u(z) = \sqrt{a^2 u(x)^2 + b^2 u(y)^2}$$

$$z = axy \text{ or } z = a\frac{x}{y} \quad \Longrightarrow \quad \frac{u(z)}{|z|} = \sqrt{\left(\frac{u(x)}{x}\right)^2 + \left(\frac{u(y)}{y}\right)^2}$$

The ratio $u(x)/|x|$ is the **relative uncertainty**.

Example 3 — Perimeter, Area, and Average Speed (*p. 10*)

A rectangular house has length (18.20 ± 0.29) m and width (11.50 ± 0.29) m. Find the perimeter and area. If one lap around the house takes (27.40 ± 0.29) s, find the average speed.

 Solution

Example 4 — Sinkhole Volume (*p. 11*)

A sinkhole on Russia's Yamal Peninsula is measured as 42.20 ± 0.29 m long, 20.30 ± 0.29 m wide, and 21.50 ± 0.29 m deep. Find its volume and uncertainty.

 Solution

Try It 2 — Gravitational Potential Energy Uncertainty

A marble has measured mass $M \pm u_1$, height $H \pm u_2$, and local gravitational acceleration $g \pm u_3$. Taking the ground as zero potential, which expression gives the uncertainty in $U = Mgh$?

(A) $u_1 + u_2 + u_3$

(B) $\frac{u_1 + u_2 + u_3}{3}$

(C) $\sqrt{u_1^2 + u_2^2 + u_3^2}$

$$(D) \sqrt{\left(\frac{u_1}{M}\right)^2 + \left(\frac{u_2}{H}\right)^2 + \left(\frac{u_3}{g}\right)^2}$$

$$(E) MgH \sqrt{\left(\frac{u_1}{M}\right)^2 + \left(\frac{u_2}{H}\right)^2 + \left(\frac{u_3}{g}\right)^2}$$

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 Solution

Core Ideas — SI Units and Dimensions

The seven SI base quantities are length (L , metre), mass (M , kilogram), time (T , second), temperature (Θ , kelvin), electric current (I , ampere), amount of substance (N , mole), and luminous intensity (J , candela). In mechanics, every dimension can be written as $M^a L^b T^c$.

Dimensions are written in square brackets, for example $[F] = MLT^{-2}$. Dimensional analysis requires both sides of an equation to have the same dimensions, and only quantities with the same dimensions may be added or subtracted.

Quantity	SI unit	Dimension
Area	m^2	L^2
Volume	m^3	L^3
Speed	m/s	LT^{-1}
Acceleration	m/s^2	LT^{-2}
Force	$\text{N} = \text{kg} \cdot \text{m/s}^2$	MLT^{-2}
Momentum	$\text{kg} \cdot \text{m/s}$	MLT^{-1}
Work / kinetic energy	$\text{J} = \text{kg} \cdot \text{m}^2/\text{s}^2$	ML^2T^{-2}
Torque	$\text{N} \cdot \text{m}$	
Angular speed	rad/s	T^{-1}
Angular momentum	$\text{m} \cdot \text{kg} \cdot \text{m/s}$	ML^2T^{-1}

Example 5 — Dimension of Planck's Constant (*p. 14*)

The photon-energy relation is $E = hf$, where f is frequency. Find the dimensions of Planck's constant h .

(A) M

(B) ML^2T^{-1}

(C) ML^2T^{-2}

(D) $ML^{-1}T$

(E) MLT^{-2}

 Solution

Example 6 — Dimensional Analysis of Momentum (*p. 14*)

Newton's general law is $F = \Delta p / \Delta t$. Suppose momentum depends on mass and speed as $p = m^x v^y$. Find x and y .

 Solution

Try It 3 — Free-Fall Speed

For an object dropped from rest through a distance h , suppose $v = cm^x h^y g^z$, where c is dimensionless. Find (x, y, z) .

(A) $(1, \frac{1}{2}, -\frac{3}{2})$

(B) $(0, \frac{1}{2}, -\frac{1}{2})$

(C) $(\frac{1}{2}, 1, -\frac{1}{2})$

(D) $(0, \frac{1}{2}, \frac{1}{2})$

(E) $(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2})$

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 Solution

Quick Check

Q1 — Interpreting a Package Label

A package is labelled “Weight: 195 ± 15 gw.” If the label follows the ISO uncertainty convention, which statement is correct?

(A) 195 is one sampled measurement.

(B) ± 15 is the standard uncertainty.

(C) ± 15 is the best estimate.

(D) The weight must vary between 180 and 210 gw.

(E) If only Type-B uncertainty is considered, the smallest scale division may be 15 gw.

 Solution

Q2 — Two Stopwatches (*select two*)

Ming and Hua measure five free-fall times with different stopwatches:

Student	Five readings (s)	Mean (s)	SD (s)
Ming	0.5893, 0.5812, 0.5868, 0.5836, 0.5874	0.5857	0.0032
Hua	0.58, 0.58, 0.58, 0.58, 0.58	0.58	0.00

Ming's stopwatch resolves 0.0001 s; Hua's resolves 0.01 s. Which statements are correct?
(*Select two.*)

- (A) Ming: 0.5857 ± 0.0029 s
- (B) Ming: 0.5857 ± 0.0015 s
- (C) Hua: 0.58 ± 0.00 s
- (D) Hua: 0.5800 ± 0.0029 s
- (E) Hua's measurement has the smaller uncertainty.

 Solution

Q3 — Relative Uncertainty of a Cube's Volume

A cuboid has dimensions 10.0 cm, 4.5 cm, and 2.0 cm, each with uncertainty 0.5 cm. What is the relative uncertainty of its volume?

(A) 1.50

(B) 0.75

(C) 0.50

(D) 0.42

(E) 0.28

 Solution

Q4 — Dimension of Pressure

Pressure is $P = F/A$. Which dimension is correct?

(A) MLT^{-2}

(B) MLT^{-1}

(C) ML^2T^{-2}

(D) $ML^{-1}T^{-2}$

(E) ML^3

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 Solution

Q5 — Dimension of a Low-Speed Drag Coefficient

At low speed, an object's air resistance is $F = cv$. What is the dimension of c ?

(A) LT^{-1}

(B) L^2MT

(C) LM^2T^{-1}

(D) MT^{-1}

(E) T^{-1}

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 Solution

Q6 — Units and Dimensions Table

Complete the SI unit and dimension for each quantity.

Quantity	SI unit	Dimension
Density D	?	
Speed v		
Work W		
Momentum p		
Force F		

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Challenge

Q1 — Sound Speed in a One-Dimensional Atomic Model

In a simplified one-dimensional metal model, atoms are masses m separated by distance d and joined by springs of elastic constant k . Suppose $v = Cm^a d^b k^c$, where C is dimensionless. Use dimensional analysis to find a , b , and c , then write v in terms of m , d , and k .

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Q2 — How Many Quantities Share a Dimension?

Among the following five quantities, how many have the same dimension?

(甲) momentum, (乙) force, (丙) kinetic energy, (丁) heat, (戊) work

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

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 Solution

Literacy-Oriented Set — Photogate Measurement of g

[Figure: A U-shaped infrared photogate apparatus measuring a steel ball in free fall.]

A photogate uses an emitter and photodiode receiver. When a falling steel ball blocks the infrared beam, the timer records the crossing time. The first gate is very close to the electromagnet, so the interval between gates can be treated as a zero-initial-speed free fall.

Q3 — Complete the Experimental Record

The four measured intervals are 0.311, 0.313, 0.311, and 0.310 s. The timer's smallest division is 0.001 s. Complete $\bar{t}$, SD , u_A , u_B , and u .

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 Solution

Q4 — Propagating Uncertainty to Gravitational Acceleration

The gate separation is $A \pm a$ m and the measured time is $B \pm b$ s. For $h = \frac{1}{2}gt^2$, $g = 2h/t^2$. Express the result and its uncertainty in terms of A , a , B , and b .

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 Solution

Exam Archive

Q1 — Jewelry Scale (2012, 112 分科)

A jeweler measures a gold ornament five times with an electronic balance whose smallest division is 1 mg. The mean is $m_{AV} = 95.367823$ g and the computed combined uncertainty is $u_C = 0.35686524$ mg. Which statement is correct?

(A) $u_A = SD$

(B) $u_B = 1$ mg

(C) $u_C = (u_A + u_B)/2$

(D) Report $m_{AV} = 95.4$ g and $u_C = 0.4$ mg.

(E) Report $m_{AV} = 95.36782$ g and $u_C = 0.36$ mg.

 Solution

Q2 — Water-Wave Speed by Dimensions (2010, 110 指考)

In a ripple tank, wave speed may depend on gravitational acceleration g , water density ρ , and depth D . Using dimensions only, which proportionality can have the dimension of speed?

(A) $v \propto gD$

(B) $v \propto \rho gD$

(C) $v \propto \sqrt{gD}$

(D) $v \propto g\sqrt{\rho D}$

(E) $v \propto 1/\sqrt{gD}$

Q3 — Dimension Matching in Periodic Motion (2009, 109 指考)

A particle executes periodic motion. The mean square displacement is X^2 , the mean square momentum is P^2 , and the total mechanical energy is E . Which expression has the dimension of a period?

(A) $\sqrt{X^2 P^2 / E}$

(B) $X^2 P^2 / E$

(C) $\sqrt{X^2 E^2 / P^2}$

(D) $E \sqrt{X^2 / P^2}$

(E) $E \sqrt{P^2 / X^2}$

Glossary

English term	中文	Definition / Note		
measurement	測量	Experimental determination of a physical quantity		
measured value	測量值	Value read or calculated from an observation		
true value	真值	Ideal value; often unknown in an actual experiment		
error	誤差	Measured value minus true value		
uncertainty	不確定度	Quantified spread or lack of certainty in a result		
Type-A evaluation	A 類評估	Statistical evaluation from repeated measurements		
Type-B evaluation	B 類評估	Evaluation from instrument resolution, reference data, or other non-statistical information		
standard deviation	標準差	Measure of the spread of a data sample		
standard uncertainty	標準不確定度	Combined uncertainty $u(X)$, usually from quadrature		
significant figures	有效數字	Digits that meaningfully represent a measured value		
relative uncertainty	相對不確定度	$u(x)/$	x	\$

uncertainty propagation	不確定度的組合／傳遞	Combining input uncertainties into a derived quantity		
quadrature (root-sum-square)	平方和原則	$\sqrt{u_1^2 + u_2^2 + \dots}$		
SI base quantity	SI 基本量	One of the seven quantities defining the International System of Units		
dimension	因次	Representation such as $M^a L^b T^c$		
dimensional analysis	因次分析	Checking or deriving relations using dimensions		
length	長度	SI base quantity; dimension L		
mass	質量	SI base quantity; dimension M		
time	時間	SI base quantity; dimension T		
force	力	$F = ma$; dimension MLT^{-2}		
momentum	動量	$p = mv$; dimension MLT^{-1}		
work	功	Force through a displacement; dimension ML^2T^{-2}		
kinetic energy	動能	Energy of motion; dimension ML^2T^{-2}		
potential energy	位能	Energy associated with position or configuration		
power	功率	Rate of doing work		

pressure	壓力	Force per area; dimension $ML^{-1}T^{-2}$		
torque	力矩	Force times lever arm; dimension ML^2T^{-2}		
angular momentum	角動量	Length times momentum; dimension ML^2T^{-1}		
photogate	光電計時器／光電閘	Infrared emitter–receiver timer used to measure motion		